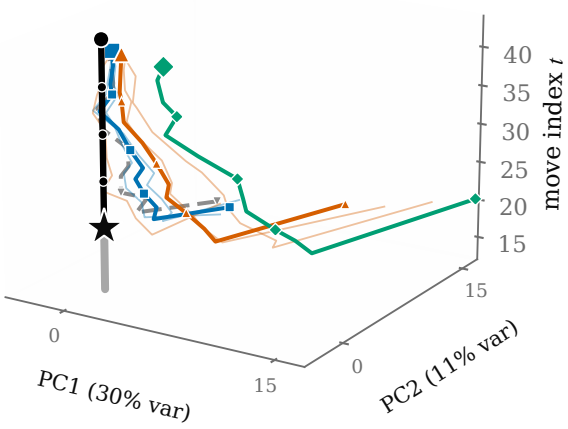
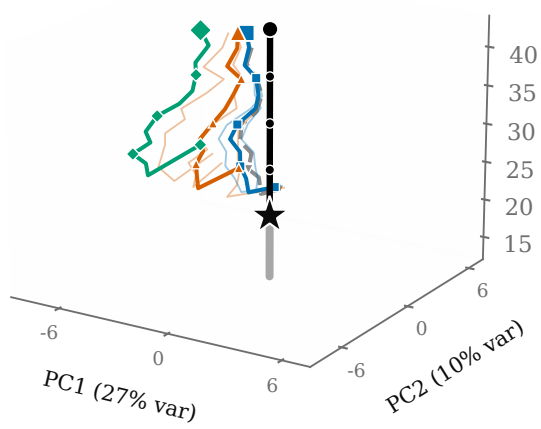
(a) where the branches go: the fork in each representation space, one path per rule

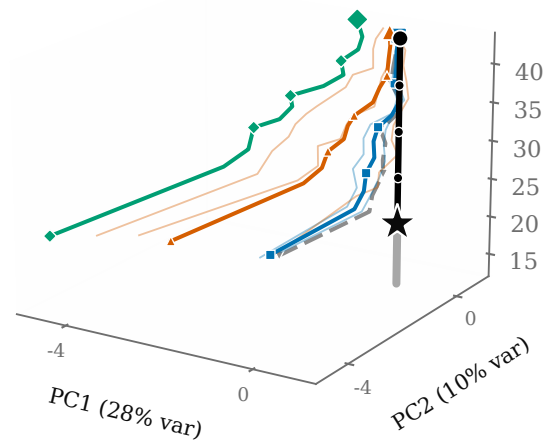
raw layer 5 (reference space)



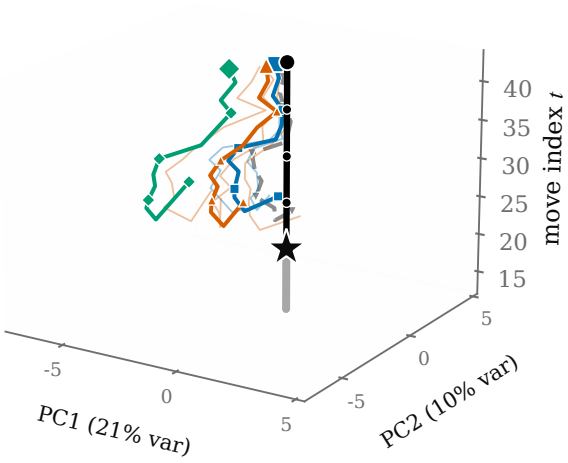
PAnneal (p-anneal, pen 0.042)



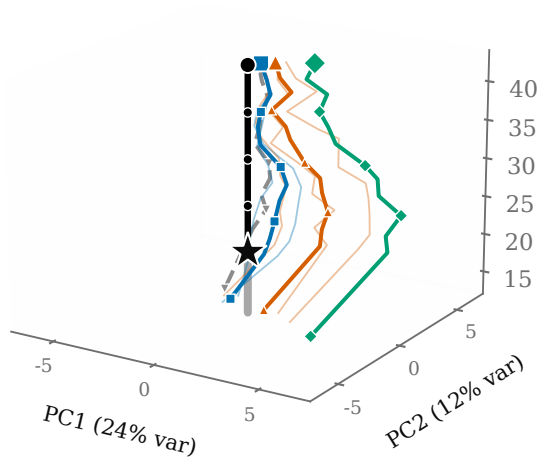
Standard (L1 0.059)



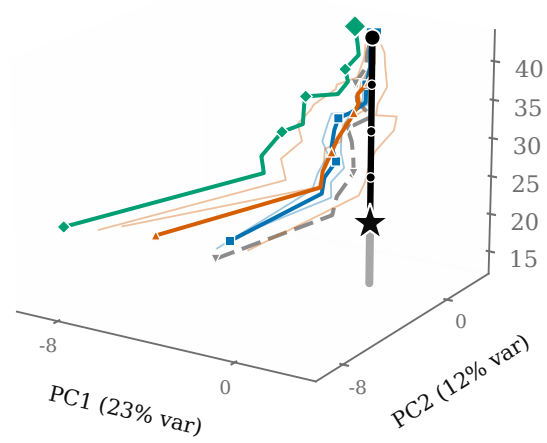
GatedSAE (L1 1.37)



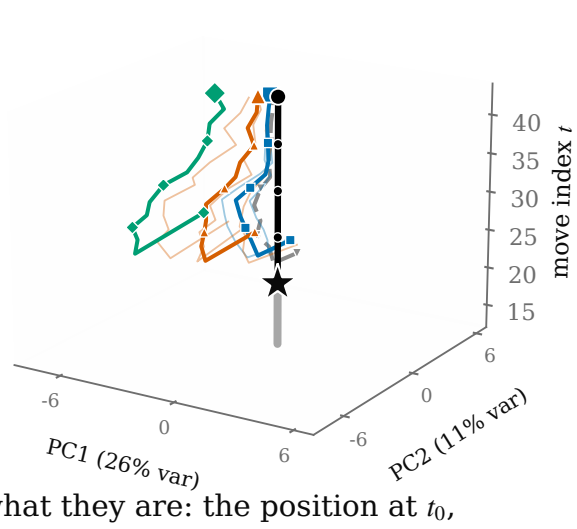
TopK (k=80)



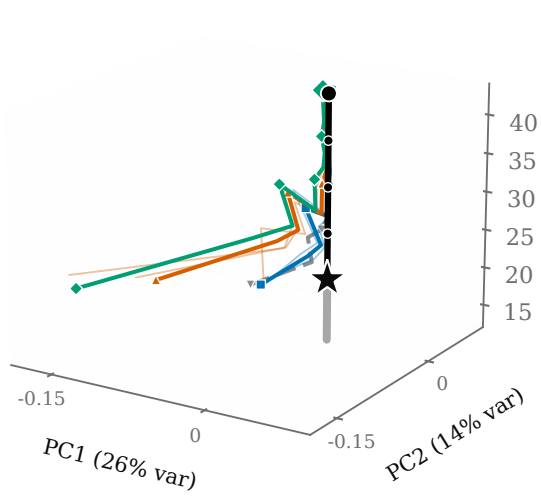
GatedAnneal (p-anneal, pen 0.955)



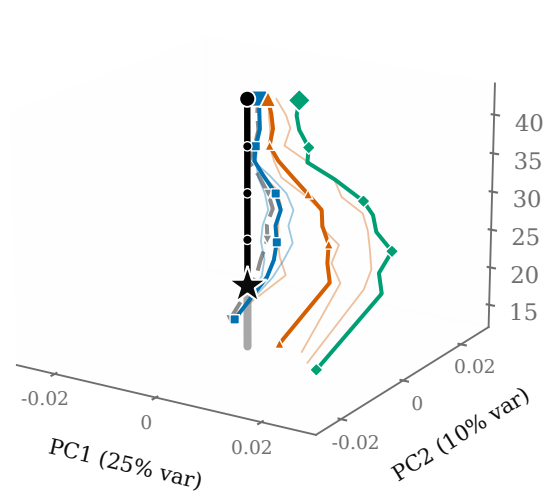
BatchTopK (k=80)



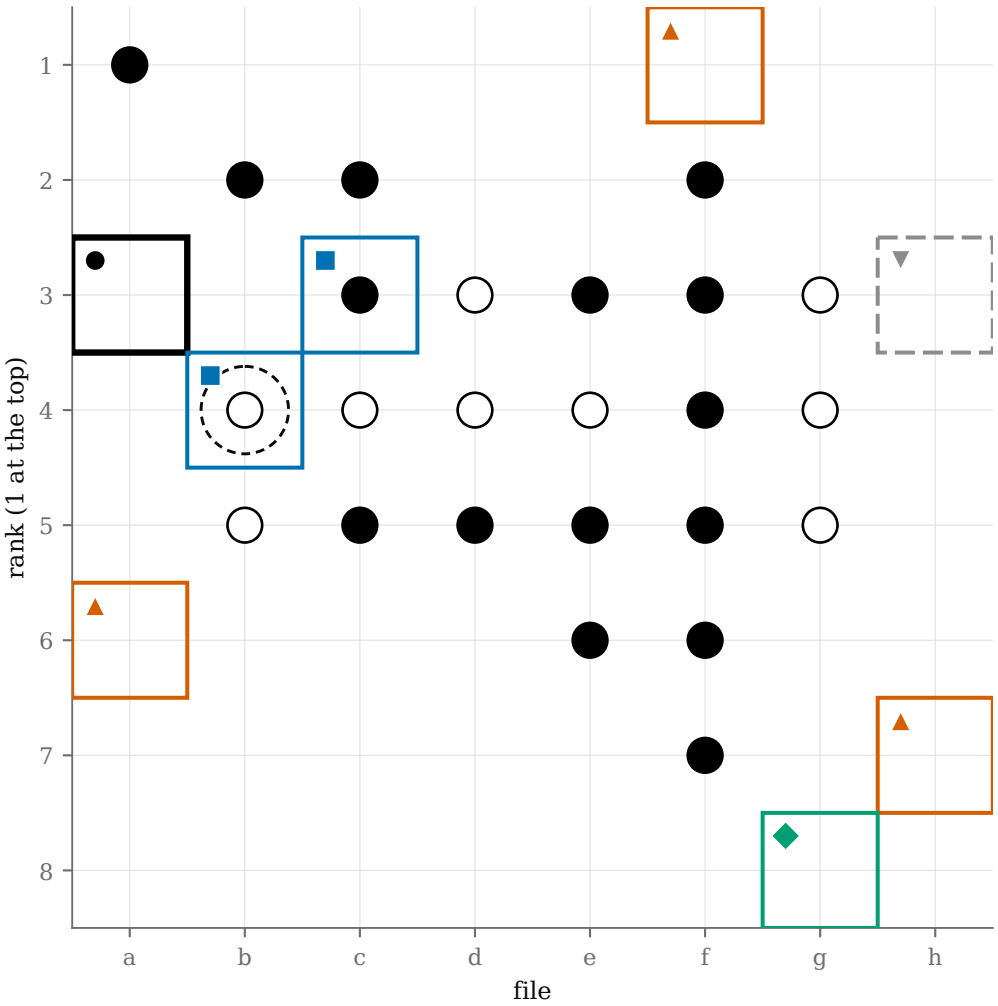
CosineSpaDE (γ=1.0)



SpaDE (λ=1e-3, γ=1e-4)



(b) what they are: the position at t_0, and the move each branch plays there



the same 8 branches in both panels, game 1235, t_0=20

(a) one path per rule: the mean of its member categories' displacements, taken in the plane (PCA is an uncentred linear map, so that equals projecting the mean displacement) over those members drawn faint. Plane refit per space — arb. units; the three ticks per plane axis are a within-panel scale only, cells are NOT comparable across the grid; view elev=18, azim=-58; every 2nd ply

(b) 10 moves are legal at t_0; the played move and one of the other 9 are branched

the grouping is weak. within-rule ÷ between-rule mean pairwise distance = 0.88 here; median 0.945 [IQR 0.872–1.020] over all 1550 branch-complete games, above 1 in 30% of them (raw layer 5, full 512-d, $t \geq t_0$). A rule's members travel barely closer to each other than to another rule's, so the faint members and not the three means carry the evidence — `python -m spade.violation_divergence.rule_groups`

one legal control is drawn, as the null. Its own null — control 2, scored against control 1 exactly as the illegal branches are — is not drawn here; it is in ../game_trajectory and in the parent README, which is where the legal-vs-legal scale lives

one game (n=1): an illustration of the design, not a claim — the study's numbers, with their CIs over 13,531 branches, are in the parent README

game 1235 is SELECTED, not a typical draw: it is the argmax, over all 1550 branch-complete games, of the mutual separation of its six illegal displacement trajectories, scored in raw layer 5 in the full 512-dimensional space and never in an SAE space or a 2-D projection. It is the clearest available illustration of the construction, not a typical one (population median 0.705, this game 1.080).

- shared prefix (branches identical)
- reference (the game as played)
- R1 placement (mean of 2)
- R2 bracketing (mean of 3)
- R3 turn (mean of 1)
- one category inside a rule (within-rule spread)
- branch point t_0
- control: another legal move (null)

discs in absolute colour, black to move at t_0; outlined cell = a branch square, its marker in the corner so a disc underneath stays visible; dashed ring = a disc the played move flips; a rule's member categories share one colour and marker here, and are told apart by their square

R1 placement — a move puts one disc of the mover's color on an empty square.
R2 bracketing — the disc must close a gapless line of opponent discs with one of the mover's own.
R3 turn — colors alternate; legality is judged for the mover, not the opponent.